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RESEARCH ARTICLE

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Blockchain and crypto forensics: Investigating crypto frauds

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Abstract

In the past few years, cryptocurrency has gained widespread acceptance because of its decentralized nature, quick and secure transactions, and potential for investment and speculation. But the increased popularity has also led to increased cryptocurrency fraud, including scams, phishing attacks, Ponzi schemes, and other criminal activities. Although there is little documentation of cryptocurrency fraud, an in-depth study is essential to recognize various scams in different cryptocurrencies. To fill this gap, a study investigated cryptocurrency-related fraud in various cryptocurrencies and provided a taxonomy of crypto-forensics and forensic blockchain. In addition, we have introduced an architecture that integrates artificial intelligence (AI) and blockchain technologies to investigate and protect against instances of cryptocurrency fraud. The suggested design's effectiveness was evaluated using several machine learning (ML) classification algorithms. The conclusion of the evaluation confirmed that the random forest (RF) classifier performed the best, delivering the highest level of accuracy, that is, 97.5%. Once the ML classifiers detect cryptocurrency fraud, the information is securely stored in the Inter-Planetary File System (IPFS); the document's hash is also stored in the blockchain using smart contracts. Law enforcement can leverage blockchain technology to secure access to fraudulent cryptographic transactions. The proposed architecture was tested for bandwidth utilization. Despite the potential benefits of blockchain and crypto-forensics, several issues and challenges remain, including privacy concerns, standardization, and difficulty identifying fraud between crypto-currencies. Finally, the paper discusses various problems and challenges in blockchain and crypto forensics to investigate crypto-currency fraud.

KEYWORDS

blockchain forensics, crypto forensics, crypto frauds, crypto scams, cryptocurrencies, smart contracts

1 | INTRODUCTION

Digital currencies or cryptocurrencies have grown rapidly in recent years. Any central body does not issue these cryptocurrencies but relies on cryptography for security.[1,2] Other key features of cryptocurrencies are

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pseudo-anonymity, the absence of middlemen, and decentralization. It is not easy to trace fund movement in routine transactions involving fiat currencies, regardless of whether the transactions were done in cash or through financial institutions. Data access is usually difficult with fiat currencies due to authorization and the difficulty of transmitting data between financial institutions concerned. It gets even more problematic when transactions incorporate several financial institutions. Most cryptocurrencies, such as Bitcoin (BTC) and Ethereum, are open-source, and all transactions are documented on a blockchain that is available to everyone. Anyone may view real-time cryptocurrency transaction data and follow fund movements on blockchains. Despite the security these digital currencies provide, all cryptocurrency usage is not necessarily legal. We must accept that they are frequently exploited as a tool for fraud, carrying out unlawful operations or money laundering.[2–4]

Transactions with cryptocurrencies are allegedly more private and subject to less regulation than transactions in traditional currencies, attracting fraudsters to utilize them to carry out their crimes. In other words, all criminals use cryptocurrencies with the same intention: to hide their illicit cash from law enforcement. Cryptocurrencies can be classified into public-oriented cryptocurrencies and private-oriented cryptocurrencies. Because every transaction on a blockchain is transparent and available to those who have access to it, and all transaction data are traceable, public-oriented cryptocurrencies do not achieve this purpose. In contrast, all transactions in private-oriented cryptocurrencies are concealed, visible only to the persons involved, and the specifics of each transaction are cryptographically secured, and untraceable.[5] Because private-oriented cryptocurrency is more difficult to track than public-oriented cryptocurrency, it enables cybercriminals to conduct unlawful transactions easily. As measured by Chainalysis,[6] the total number of transactions across all cryptocurrencies increased to $15.8 trillion in 2021, 567% high from the totals in 2020. In 2022, there was a noticeable surge in the adoption of cryptocurrencies for gambling and betting, driven by their increasing acceptance among both businesses and individuals. During the same year, the volume of illicit cryptocurrency transactions reached a historic peak. According to Chainanalysis research, illicit transaction volume marked its second consecutive yearly increase, reaching a staggering $20.6 billion in 2022. Furthermore, the proportion of cryptocurrency activity linked to illicit endeavors witnessed its first rise since 2019, climbing from 0.12% in 2021 to 0.24% in 2022.[7]

Forensics science involves gathering and analyzing data to draw conclusions about the fraud cases being investigated. The area of forensic science known as "blockchain forensics" and "crypto forensics" is dedicated to finding, examining, and analyzing blockchain data that is connected to cryptocurrency crime. Forensic accountants and blockchain and digital forensic specialists are well-positioned to aid in tracking crypto assets, providing clarity, and conducting a complete investigation due to their unique skill set, expertise, and continued education.

1.1 | Comparisons with the existing surveys

As more people become involved in the new digital monetary system, such as BTC, the vulnerabilities inherent in these transactions become clear. Due to pseudo-anonymity's anonymity and privacy, cryptocurrencies have become a target for cybercriminals. It is challenging to identify cybercrime-related activity. Researchers have worked hard to analyze and design novel defensive measures to avoid these behaviors. The use of crypto forensics and blockchain forensics has been investigated for detecting and mitigating crypto fraud. Through various evaluations, an understanding of the current situation has been established, and a roadmap for further research has been laid out. Inspired by these findings, we studied numerous survey papers for the proposed survey. This section explores cutting-edge efforts on crypto scams or frauds, blockchain forensics, and their integration into investigating frauds. Table 1 compares prevailing surveys to planned ones.

1.2 | MOTIVATIONS AND CONTRIBUTIONS

1.2.1 | Motivations

The motivation for the present work is described as follows:

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TABLE 1 Relative comparison of the proposed survey with the state-of-the-art surveys.

Related survey	Year	Objective	Key contributions	Limitations and open issues
Soni and Maheshwari[8]	2018	Examine various attacks in the BTC system.	This study examines protocol security.	To examine the crypto scams other than the BTC network no qualitative approaches were used.
Lee et al.[9]	2019	In-depth investigation on how criminals exploit cryptocurrencies on the Dark Web.	Built a novel framework MFScope that collects data from dark websites and discovered that 80% of the cryptocurrency addresses are being utilized for criminal reasons.	Does not uncover any apparent patterns of activity on cryptocurrency exchanges.
Nikkel[10]	2020	Introduces Fintech forensics that deals with financial technology for the investigation and reconstruction of unlawful financial behavior.	Fintech Forensics will train Fintech systems as well as financial knowledge of payment systems. When financial technology is incorporated, this will assist society in fighting crime more effectively.	Does not clarify how effectively existing digital forensic investigation methodologies apply to financial technology that involves the exchange of funds.
Ilker and Aydos[11]	2020	Focused on crypto-ransomware and examined a forensic investigation of this assault.	Proposed a method for detecting and analyzing crypto-ransomware.	This study focuses solely on the crypto-ransomware attack.
Srivasthav et al.[12]	2021	Focused on blockchain forensic tools.	Compared the different assessed tools	The tools were not analyzed using any qualitative methods.
Bartoletti et al.[13]	2021	A comprehensive examination of cryptocurrency scams.	Created an open-source tool for scam classification.	Should be used in conjunction with additional tools such as ML techniques that measure the lifetime and economic impact of various frauds.
Mas'ud et al.[14]	2021	Examine how digital forensic investigation procedures have been applied to blockchain and cryptocurrencies.	Different phases of digital forensic investigation methods are explored.	Frameworks for blockchain and crypto forensics should be examined more.
Mashatan et al.[15]	2022	Examine how customer confidence in crypto payments is built.	Discovered the major indicators of customer trust in crypto-payment.	The study might look at the impact of client gender, expertise, and other possibly important elements like technology preparedness or contextual factors.
The proposed survey	2023	Study on blockchain Forensics and crypto forensics for investigating crypto frauds.	Investigated various crypto frauds by merging the features of blockchain forensics and crypto forensics.	NA

- The present crypto market is constantly changing, with new technologies and fraudulent schemes frequently emerging. Therefore, previous studies may not accurately reflect the industry's current state, and investigators need to stay updated on the latest trends and techniques like crypto forensics and blockchain forensics.

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- Previous state-of-the-art approaches for investigating crypto fraud have primarily focused on BTC-related frauds while disregarding other cryptocurrencies.[8] However, as the crypto market continues to evolve, it is becoming increasingly necessary for investigators to expand their knowledge and expertise beyond BTC and become familiar with other cryptocurrencies as well. This highlights the need to know the different types of crypto frauds across various cryptocurrencies.
- Despite the growing importance of Blockchain Forensics and Crypto Forensics in investigating crypto fraud, there are still limited empirical studies in this area. As a result, investigators may not be fully equipped to handle these cases without a more comprehensive understanding of the tools and techniques used in investigating crypto fraud. Therefore, a comprehensive survey can provide valuable insights into the field and help investigators more effectively combat crypto fraud.
- There has been no discussion in previous studies about the taxonomy of crypto forensics, which helps investigate crypto frauds. Previous item studies have not included a discussion of a case study to ensure fraudulent data privacy from intruders attempting to manipulate or misuse the data.
- Our research has been motivated by these limitations and aims to bridge the gap by thoroughly investigating various types of crypto frauds and applying Blockchain Forensics and Crypto Forensics to investigate them in real-world scenarios. We hope our findings can help investigators better understand the complexities of the crypto market and be better equipped to investigate and prosecute crypto fraud cases.

1.2.2 | Contributions

The considerable contributions of the present research work are as follows:

- We thoroughly analyzed the many cryptocurrency-related frauds that occur across different cryptocurrencies.
- To comprehensively evaluate cryptocurrency-related scams, we offer a taxonomy of crypto forensics.
- We conducted a case study that leveraged blockchain and AI technology to detect cryptocurrency fraud.
- Finally, we discussed many issues and challenges in blockchain and crypto forensics to investigate cryptocurrency fraud.

1.3 | Methods and materials

To ensure a comprehensive analysis of the use of crypto and blockchain forensics for detecting crypto fraud, an in-depth examination and investigation was carried out as a prerequisite for this study. We started the investigation by recommending that review papers on crypto frauds and scams be examined from recognized conferences and databases like IEEE Explore, Scopus, Google Scholar, ACM, and Science Direct. We then investigated how crypto and blockchain forensics investigate or reduce crypto fraud. The search terms crypto frauds, crypto scams, crypto forensics, Blockchain forensics, and the application of crypto forensics and blockchain forensics in detecting frauds are used in this criterion. Afterward, we analyzed the summaries and findings of the gathered papers to identify the appropriate approaches and concepts for the planned survey.

1.3.1 | Research questions

The final goal of an article is reflected in the research questions, which influence the decision-making process of primary research. This study aims to look at various crypto frauds and how blockchain analysis and crypto forensics tools might help combat these crypto scams.

RQ1: What role does cryptocurrency play in fraudulent schemes?
RQ2: How might blockchain or crypto forensics help in cryptocurrency crime investigation?

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1.3.2 | Data sources

We searched peer-reviewed journal databases such as IEEE Xplore, ScienceDirect, Scopus, ACM DL, and Google Scholar to uncover contemporary literary studies and material online. We also used additional materials, such as technical literature about the case, reports, websites, and online blogs, about the present survey.

1.3.3 | Search string

Specific keywords and their substitutes are chosen to build the search string within the defined research topics. The keywords are then placed in a particular order with the criteria "AND" and "OR" to generate the following query:

((*"Cryptocurrency frauds"* OR *"Cryptocurrency scams"* OR *"investigating methods of crypto frauds"* AND (*"Blockchain Technology Forensics"* OR *"Blockchain Technology Forensics Tools"*) AND (*"Crypto Forensics"* OR *"Crypto Forensics Tools"*) AND (*"Role of Forensics tools in investigating Frauds"* OR *"Role of Forensics tools in reducing Frauds"*))

1.3.4 | Inclusion criteria

The prerequisites for inclusion are listed as follows:

- Papers delivered at conferences, conference proceedings, or published in peer-reviewed journals or as book chapters.
- The papers concentrate on cryptocurrency frauds/scams.
- Papers focusing on Blockchain or Crypto forensics
- Papers examine how Blockchain forensics or Crypto forensics analyze cryptocurrency fraud.
- The most recent papers are given preference.

1.3.5 | Exclusion criteria

The following are the exclusion conditions.

- Papers published before 2008.
- Articles are written in a language that's not English.
- Research papers that do not specifically address cryptocurrency fraud.

In Figure 1, we depict the steps we followed to identify the most relevant research papers for our study. Initially, we retrieved 433 papers from the database, and each of these papers underwent a thorough review. Out of this initial collection, we carefully selected 101 papers that directly related to our research subject. Subsequently, we conducted a detailed analysis of these selected papers to ensure they aligned with the scope and objectives of our study. This multi-stage selection process was employed to guarantee that the papers included in our study met our specific criteria and were a suitable fit for our study's purposes.

1.4 | Structure of this survey

The current research work is organized as follows. Section 2 discusses various cryptocurrency-related frauds. Sections 3 and 4 cover the taxonomy of Crypto forensics and Blockchain forensics, respectively. Section 5 elaborates on the case study of AI and Blockchain-based framework for crypto fraud detection. Section 6 provides the various challenges in investigating crypto fraud. Finally, Section 7 concludes the paper.

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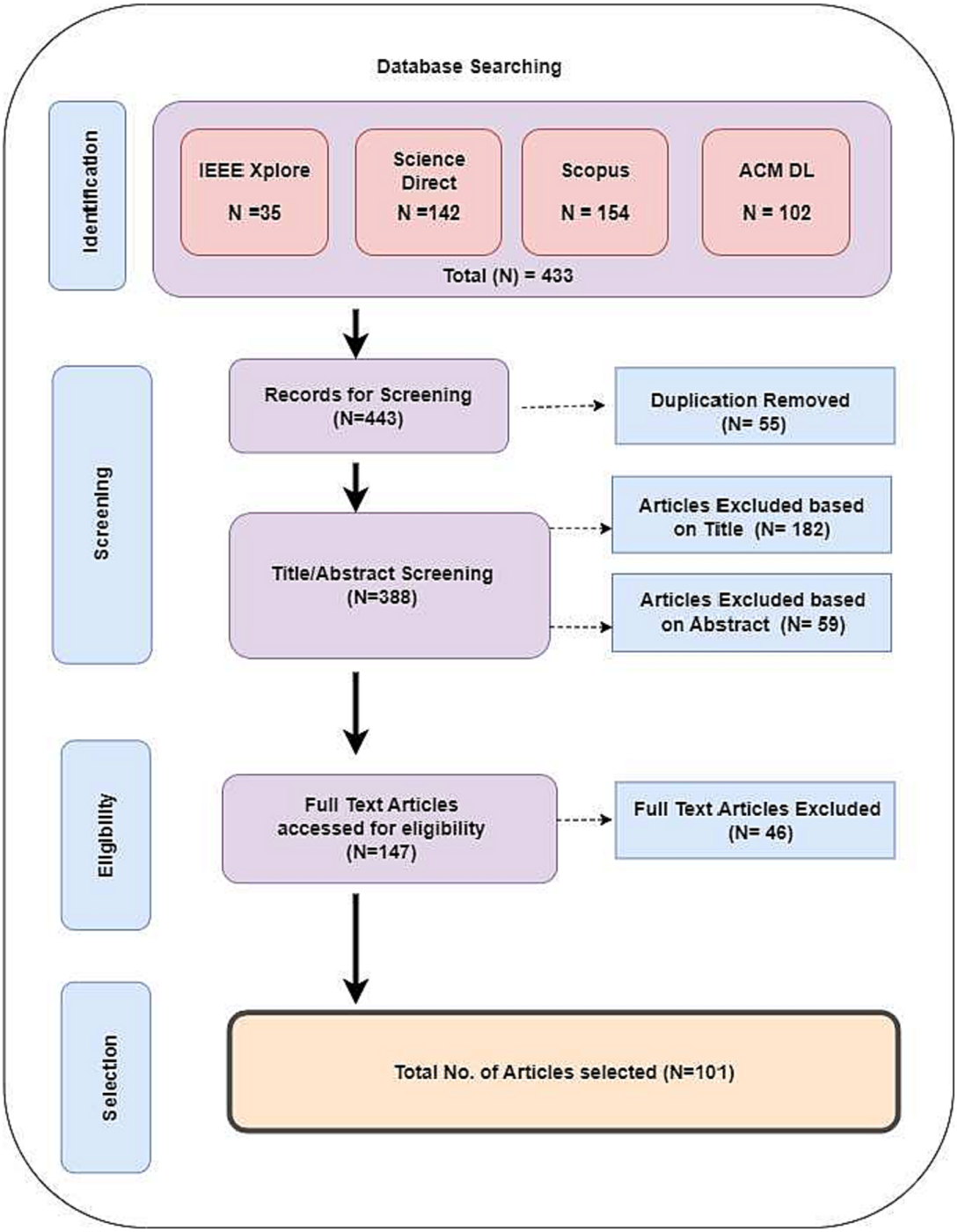


FIGURE 1 Survey criteria (N = no. of papers)

2 | BACKGROUND AND PRELIMINARIES

Scammers are drawn to cryptocurrency for a variety of reasons. There is no bank or other centralized body to identify questionable transactions and try to prevent fraud from occurring. Crypto transactions cannot be reversed; once the money is gone, it cannot be recovered. Foley et al[16] investigated the extent of cryptocurrency usage and concluded that around 46% of BTC transactions aid illicit behavior. Hackers utilize a variety of tactics to take cryptocurrencies from investment accounts, and while investors are pretty confident in the protection afforded by blockchain, hackers steal money on a daily basis. Cryptocurrency exchanges are prone to cyberattacks and have already had several intrusions in their systems. These hacks have the ability to destabilize an exchange severely, as was done with Mt. Gox. Mt. Gox was

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hacked not once, but twice.[17] The first hack occurred in 2011, and the second in 2014. Many exchanges, including Mt. Gox, have been severely hacked in an attempt to steal cryptocurrencies traded on the market. In February 2022, the BTC exchange company Wormhole suffered a devastating blow when it lost a staggering sum of $320 million due to a cyber-attack. However, this unfortunate incident is just one of many that have plagued the world of cryptocurrency since 2021. According to the Federal Trade Commission's investigation, BTC fraudsters have already stolen over $1 billion in this relatively short period. The investigation also revealed that BTC was used as the payment method in 24% of all dollar losses reported in FTC fraud reports between January 1, 2021 and March 31, 2022.

Cryptocurrency crimes are also linked to the online sale of unlawful items. The majority of these crimes take place on the darknet's secret services. It is not accessible using a standard search engine. TOR masks the IP address of visitors.[18] As a result, it is challenging for law enforcement to track down criminals. The potential use of BTC in terrorist actions is also relatively high. A message posted on a Syrian jihadist website revealed their intention to accept BTC donations. Additionally, Indonesian authorities have reported that extremist groups use BTC and PayPal to fund terrorist activities within their borders. According to Bloomberg, Jason,[19] there are four primary types of cryptocurrency-related crimes: money laundering, illicit transactions, tax evasion, and extortion. BTC exists as a string of digits called the "BTC address." Owners may store this address in a physical wallet for added security or use cloud-based wallet services like Coinbase. However, relying on such services can increase the risk of BTC theft, as a third party stores private keys. Hackers can gain access to these keys and steal BTC from clients. Virtual currency theft through online wallets has become common, as almost every theft incident involves such wallets.

The various Crypto-related frauds, as shown in Figure 2, are detailed below.

2.1 | Investment schemes

In these schemes, scammers approach investors claiming to be experienced "investment managers." As a component of their deceitful scheme, the alleged investment managers proclaim that they have made millions of dollars through cryptocurrency and assure their targets that they can also reap profits from similar investments. However, to initiate the process, the fraudsters demand a fee. Rather than generating returns, the scammers steal the initial payments. Moreover, under the guise of facilitating payment transfers and cryptocurrency access, the culprits may solicit personal identifying information from their victims.

Figure 3 shows the amount of cryptocurrency lost by reporting crypto frauds during recent years. The amount of investment-related scams is the highest.

- High Yield Investment Programs (HYIPs): These are the unauthorized investment opportunities typically run by unlicensed individuals. The commitment to high returns with small or no risk to the investor is characteristic of an HYIP scam. An HYIP website may guarantee yearly (or monthly, weekly, or daily!) returns of 30% or 40%—or more. Some of these schemes may refer to a "prime bank" initiative. HYIPs frequently use cryptocurrency assets and rely on existing investors to bring new ones. J. Neisius and R. Clayton[21] examined the relationship between aggregators and

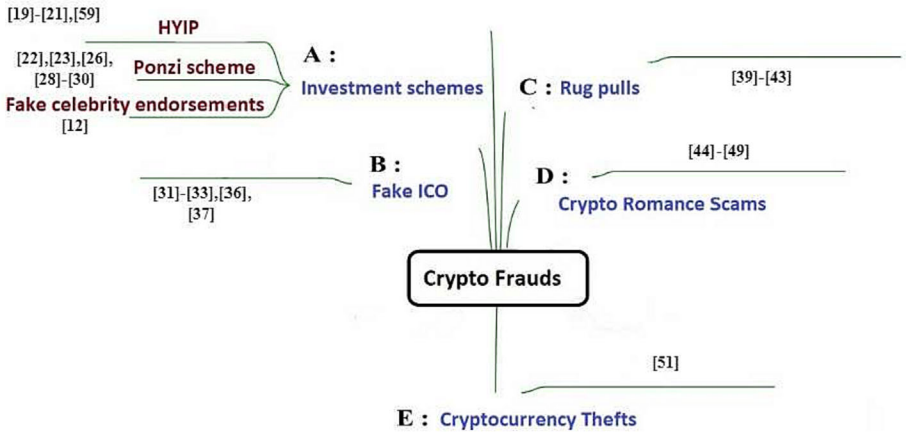


FIGURE 2 Types of various crypto frauds.

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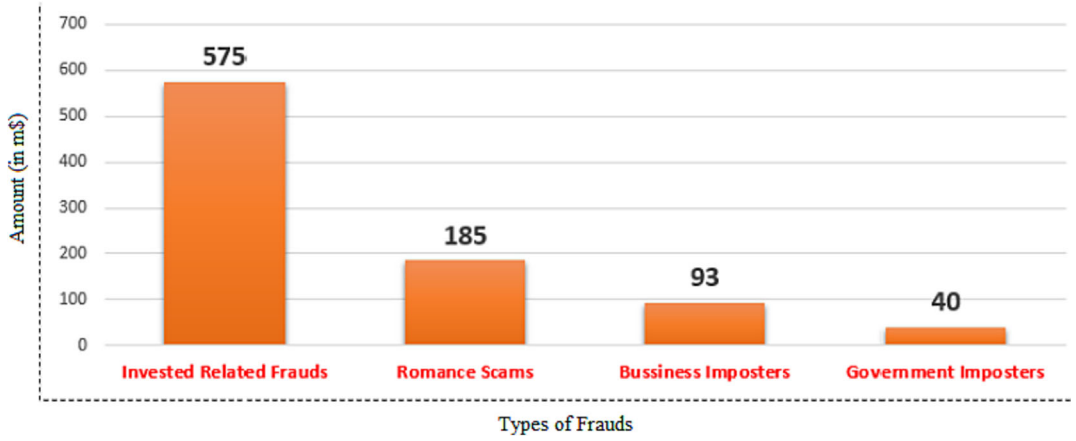


FIGURE 3 Amount of cryptocurrency lost by report crypto scams (Jan 2021 to March 2022).[20]

HYIPs. They disclosed selective payments, that is, HYIPs that pay only to aggregators or selected investors, defrauding innocent investors. Toyoda et al[22] described a clustering algorithm for detecting relationships between HYIP websites. They used HTML elements and file directory structures to connect different websites and show that the same person owned them. Agarwal et al[23] focus on understanding HYIPs and the numerous causes for their existence in today's environment.

• *Ponzi scheme:*The present scheme constitutes an investment scam that involves dispensing alleged profits to existing investors from funds obtained from new investors. Promising significant returns with minimal or no risk, organizers of Ponzi schemes often entice new investors. However, instead of engaging in legitimate investment endeavors, fraudulent operators in numerous Ponzi schemes concentrate on accumulating fresh funds to fulfill payouts pledged to earlier investors while diverting some of these "invested" funds for their advantage. Examples of Ponzi schemes are as follows:

- *GainBTC Scam:* This scam was designed to attract new investors unfamiliar with the crypto industry. In this specific fraud, seven employees reported to a person dubbed the "Seven Stars," who were in charge of bringing in new people under the GainBTC banner. The GainBTC Ponzi scheme attracted new investors by promising a 10% monthly return on their investment for the next 18 months. The GainBTC fraud may have cost over Rs 1 lakh crore (Trillion) to over 1 lakh victims.
- *BitConnect:* This scheme was founded in 2016 as a BTC lending service that promised 40% monthly returns. BitConnect was a website where investors could lend their cryptocurrencies to the firm in exchange for large profits based on the loan length. Because of its multilevel marketing structure and absurdly high rewards (1% daily compounded interest), Bitconnect was accused of being a Ponzi scam. Bitconnect was revealed in 2018, eventually collapsing and costing its investors $3.5 billion.
- *OneCoin:* It was a Ponzi scam presented as a cryptocurrency by OneCoin Ltd (registered in Dubai) and OneLife Network Ltd, which had offices in Bulgaria (registered in Belize). Although the company offered instructional resources, such as educational materials concerning cryptocurrency, the products were part of a multilevel marketing (MLM) initiative. Participants who bought course packages were incentivized to bring in more individuals, with rewards awaiting those who succeeded. According to the program's terms, buyers of these courses would receive tokens to be utilized in mining OneCoins. Much of the course content was reported to be plagiarized. From 2014 to 2016, it brought in $4 billion.
- *PlusToken:* Most of the scam's promotion was done through the Chinese messaging app WeChat, tempting investors with the promise of 10-30% monthly returns on investment. PlusToken drew nearly three million investors, mainly from China, South Korea, and Japan. Instead, PlusToken defrauded investors until 2019, when it was shut down, losing investors more than $3 billion.

Chen, Weili, et al[24,25] presented an ML and data mining strategy for detecting smart Ponzi scams on the blockchain.

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• Fake celebrity endorsements: This particular form of investment fraud differs from the usual. To create the illusion of a prominent financial gain endorsed by a celebrity, perpetrators incorporate authentic photographs of the individual into fake accounts, advertisements, or articles. These assertions stem from authentic sources, utilizing established brand names and exhibiting professional websites and logos. However, the claimed endorsement is nothing but a fake.

Table 2 shows the relative comparison of state-of-the-art frauds on investment schemes.

2.2 | Fake initial coin offerings (ICO)

The next cryptocurrency scam perpetrated by scammers is called the Fake ICO. The acronym ICO stands for Initial Coin Offerings. These are regarded as "common fundraising tactics" businesses and startups employ. The cryptocurrency industry's counterpart to an IPO is an ICO. However, the main difference is that an IPO is often for well-established enterprises, whereas an ICO is typically for new and dangerous ones. Regarding fake ICOs, several frauds involving businesses vanished after receiving substantial sums of money. The most well-known instance of a fake ICO

TABLE 2 Relative comparison of state-of-the-art crypto frauds on investment schemes.

Paper	Year	Key findings	Limitations and open issues
Song and Kong[26]	2022	Introduced a method for determining whether or not a contract is a smart Ponzi scheme by deploying two scenarios–contracts and long-term contracts.	The fund's functioning status is monitored using only three indicators. There will be more indications applied. The proposed approach is not completely automated.
He et al.[27]	2022	Provided a mechanism for noticing Ponzi scheme contracts on the Ethereum-based blockchain network.	It is tough to evaluate the effectiveness of the proposed system for detecting all types of Ethereum Ponzi contracts. Other approaches should be associated with it.
Jin et al.[28]	2022	Proposed a Ponzi-Warning method for Ethereum Ponzi schemes that conducts feature extraction and fusion at both the transactional and code level	The proposed technique of transaction-aware channel for enhanced Ponzi detection needs to be improved. It should be supplemented with some extra tactics.
Fan et al.[29]	2021	Recognized smart Ponzi schemes on the blockchain and also introduced a model based on anti-leakage smart Ponzi schemes.	The suggested system cannot detect other crypto frauds and contract privacy data theft in blockchain networks in real time.
Wang et al.[30]	2021	Proposed a Ponzi scheme detection technique for smart contracts based on LSTM.	The projected technique of Ponzi scheme smart contracts detection needs to be improved. It should be supplemented with some other methods in Deep learning.
Chen et al.[31]	2021	A semantic-aware detection technique (SADPonzi) for detecting Ponzi scams in Ethereum-based smart contracts was proposed	The suggested approach cannot identify crypto frauds in blockchain networks other than Ethereum smart contracts in real-time.
Yu et al.[32]	2021	Proposed a detection model built on GCN to discriminate Ponzi contracts properly.	The proposed Ponzi scheme detection technique cannot detect fraud on blockchain networks in real-time.
Chen et al.[24]	2019	Presented a classification model to discover latent smart Ponzi schemes.	The proposed method does not examine each smart contract to provide early warning of scams, and the classification model employed in this approach needs to be more accurate.
Chen et al.[25]	2018	Introduced a method for detecting Ponzi schemes leveraging data mining and ML technologies.	The classification model employed in the proposed detection strategy of Ponzi schemes is not particularly effective and cannot detect frauds in blockchain networks other than Ethereum smart contracts in real time.

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is PlexCoin. This fictitious cryptocurrency lured high-end investors into making sizable investments in a fake cryptocurrency by promising shockingly huge returns (1300%) within the first 30 days. Investment in fake cryptocurrencies occurred in 2020, largely during the lockdown period. Individuals who fell prey to this fraudulent scheme had procured the "Morris Coin" from the cryptocurrency exchange known as Franc Platform, located in Coimbatore. This digital asset was marketed at Rs 15,000 for a package of ten coins, and buyers were subjected to a mandatory lock-in period spanning 300 days. The depositors were handed an e-wallet and informed that the coin's worth would skyrocket once it was traded. However, the coin's founders diverted the funds and unlawfully financed certain real estate assets without indicating any income source.

Basu, Saunak, et al[39] suggested that human specialists can use AI agents as a filter that rates poorly and spend more time researching projects. Karimov et al[38] investigated how effectively one can anticipate if an offering is a fraud. A variety of linear and nonlinear classification algorithms have been employed to identify the key predictors of an offering's failure and gain insights into the underlying relationships. The aim is to identify the most critical factors contributing to an offering's failure and interpret the relationships among them. It explained that using ex-ante characteristics, one can anticipate a scam with an accuracy of 65%–70% and that nonlinear ML models outperform classical logistic regression. According to their findings, extreme gradient boosting (XGBoost) was the best method for identifying fraudulent ICOs. Zheng et al[35] solved the problem of imbalanced classes by removing linguistic features from the ICO white paper and trained the prediction model with several ML and deep learning algorithms. Yan, Siyuzhuo[34] showed that BP neural network had a high accuracy in determining the legality of the ICO. Chiu, Tiffany, et al[33] suggested that Linguistic Inquiry and Word Count (LIWC) might help uncover probable ICO fraud situations. Table 3 shows the comparative comparison of state-of-the-art frauds on ICO schemes.

2.2.1 | Pump and dump schemes or rug pulls

The objective behind this scheme is to symbolically pull the rug out from under investors, causing their investments to crumble. The fundamental concept of this fraudulent activity involves disseminating false information to deceive

TABLE 3 Relative comparison of state-of-the-art crypto frauds on ICO scheme.

Paper	Year	Key findings	Limitations and open issues
Chiu et al.[33]	2022	Used textual analysis methods like Linguistic Inquiry and Word Count (LIWC) to uncover probable ICO fraud situations.	The suggested approach is incapable of identifying various crypto frauds in real time.
Yan[34]	2022	Focused on the ICO Economic Crime Model, constructed on the BP Neural Network.	The recommended ICO Economic Crime Model could be more effective when spotting different cryptocurrency scams in real-time.
Zheng and Wang[35]	2022	Solved the problem of imbalanced classes by extracting linguistic features and training the prediction model with ML and deep learning algorithms.	The method for locating fake ICO programs that have been suggested needs to be revised. Additional strategies need to be added to it.
Hornuf et al.[36]	2022	Examined the material provided before the ICO and proposed a method that indicates the possibility of fraud and documents the magnitude of the occurrence.	The suggested method for real-time detection of initial coin offering fraud across many blockchain networks is insufficient.
Swartz[37]	2022	Provide an approachable, non-technical introduction to ICOs frauds and the 2017 bubble.	It is necessary to investigate more.
Karimov and Wójcik[38]	2021	Several linear and nonlinear classification algorithms are used to recognize the important predictors of fraudulent ICOs.	The proposed technique must be developed to discover trustworthy ICO proposals in real time.
Basu et al.[39]	2021	This study suggests that humans deploy AI agents as filters and pay greater attention to the riskier suggestions because they have an asymmetric framing impact on the recommendations.	The proposed method needs to be revised to detect crypto crimes in real time accurately. It should be investigated more.

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individuals into thinking that a particular investment holds substantial potential. The scammer patiently awaits the moment when the Value of the investment skyrockets to an exorbitant level, only to sell off the number of stocks or assets they had initially purchased. The perpetrator then escapes with the proceeds obtained from unsuspecting victims. A noteworthy example of this type of scam was the emergence of the $SQUID Game coin, named after the popular Netflix series Squid Game, in the past year. The coin had nothing to do with the show, but that did not stop it from creating a buzz. It quickly jumped from just over a penny to $2800. After hitting this peak, it dropped to a few cents a few minutes later.

Cryptocurrency may provide a new variant on the classic pump and dump scheme, in which stockholders attempt to push up the Value before dumping their shares at an artificial peak. This is frequent in the crypto sector during the ICO stage and even after, when misleading promises may boost demand and allow the cryptocurrency's originators or majority holders to gain enormous bogus profits. According to Li et al,[40] pump-and-dump actions cause short-term bubbles with rapid price, volume, and volatility increases. Prices reach a high in minutes, then quickly reverse. Unlike Li et al.,[40] Hamrick, J.T.,et al.[41] examined long-term (so-called goal-based) pump-and-dump strategies in which pump signals do not achieve a buy objective for days or even weeks. Nizzoli, Leonardo, et al[42] examined social media networks and discovered that 56% of Telegram messages were sent by bots or prohibited accounts, while 93% of Twitter bot tweets were sent to Telegram, supporting both "pump and dump" and Ponzi schemes. Nghiem et al[43] studied "pump-and-dump" strategies in the cryptocurrency market and predicted the identification of a "pump-and-dump" target value and estimated the maximum trading price with an error margin of 6.1%. They also created a model to access the high price using a neural network-based algorithm based on market and social media signals.

Rug pulls are a common technique used in DEXs, the DeFi version of market exchanges, notably in Uniswap, in which fraudulent agents create an ERC-20 (Ethereum Request for Comments 20) token and list it on a DEX, pairing it with a dominant cryptocurrency like as USD or Ether. Uniswap is an Ethereum blockchain on-chain liquidity protocol that enables trustless token swaps, meaning that all transactions are conducted via smart contracts without an intermediate or trusted party. Mazorra et al[45] investigated the DEX thefts and scams and offered an ML system to classify tokens as frauds. It gives a conceptual arrangement of three main forms of rug pulls: simple, sell, and trap-door rug pulls. Table 4 gives the relative comparison of state-of-the-art frauds on Rug pull schemes.

2.2.2 | Crypto romance scams

In this scam, fraudsters frequently make the target person's trust initially through affectionate conversations on dating apps or other social media. Then start offering them crypto 'advice' on where to put their money. Scammers may use

TABLE 4 Relative comparison of state-of-the-art crypto frauds on rug pulls scheme.

Paper	Year	Key findings	Limitations and open issues
Chadalapaka et al.[44]	2022	Described the two deep learning techniques for detecting Pump and Dump strategies in altcoins.	Because of the volatility of cryptocurrencies, the suggested approach is ineffective for detecting crypto pump and dump in real-time.
Mazorra et al.[45]	2022	Focused on thefts and scams in the DEX and introduced methods based on ML to designate tokens as frauds.	The tokens' source code needs to be more accurate, and the study focused solely on decentralized exchanges.
Nghiem et al.[43]	2021	Identified the coin under pressure using a neural network-based algorithm and then created a model to assess the price high.	The suggested research should examine pump-and-dump scams on other exchanges, including Yobit and Bittrex.
Li et al.[40]	2021	Identified that most pump-and-dump strategies result in short-term bubbles in which values, demand, and liquidity rise, followed by a sharp reverse.	The proposed study should look at other real-time ways to avoid pump-and-dump fraud.
Hamrick et al.[41]	2021	Long-term or goal-based pump-and-dump methods were investigated.	The proposed method could be more effective at detecting crypto pumps and dumps in real time. It should be explored further.
Nizzoli et al.[42]	2020	Examined the online cryptocurrency ecosystem across numerous platforms.	It should be explored further to investigate and prevent crypto rug pull scams.

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fraudulent or false websites, cryptocurrency wallets, or even a fake/fraudulent person. The scammers then prefer to vanish with the target victim's money. Most cryptocurrency dating scams follow a pattern known as "pig butchering," in which fraudsters repeatedly flatter and make their victims feel good before cheating them, much like a farmer fattens a pig before slaughter. People in the United States reportedly lost about $185 million in cryptocurrency through romance in just 15 months, from January 2021 to March 2022.

Dealing with crypto romance scams is difficult since such frauds are rarely undertaken in large-scale drives and are not created repeatedly. As a result, they cannot be recognized by spam filtering algorithms that employ similarity detection. Dating websites are supposed to connect individuals and meet new people, rendering the idea of unwanted messages. Romance fraudsters will send personalized emails to build their victims' confidence. These exchanges closely mimic those sent between actual dating service members. In many situations, the real fraud occurs after a few weeks or months of contact has switched to other, unsupervised media. As a result, it is critical to spot romance fraudsters before they attack. Rege[46] investigated how cyber-criminals plan and conduct online romance frauds. Using a socio-organizational level of analysis, this study showed a preliminary evaluation of online criminal structure. It investigated how digital surroundings influenced cyber-criminal organizations. Whitty and Buchanan[47] investigated the emotional influence of the online dating romance scam. The findings found that victims felt humiliation, embarrassment, astonishment, anger, anxiety and stress, terror, and the sensation of being psychologically raped. Victims of online romance scams also had a poor sense of self-worth, low self-confidence, less motivation to be social, and lost trust in others.

Buchanan et al[48] conducted interviews with romance fraud victims to uncover a cognitive disagreement in which victims pay attention to some aspects but ignore those that contradict their perspective of the growing relationship. Cross et al[49] looked into the connections between romance fraud and domestic exploitation and violence through interviews with victims, revealing various similarities. G. Suarez-Tangil et al[50] introduced an ML technique for automatically detecting online dating fraud. In this, authors employed text classification and picture analysis methods to extract meaningful information from a vast data set of online dating user-profiles and identified fraudster profiles automatically. He, Xinlei, et al[51] proposed a system called DatingSec, which leveraged LSTM and user-generated textual content for noticing malicious accounts in dating apps. Table 5 describes the relative comparison of state-of-the-art frauds on Crypto Romance scams.

2.2.3 | Cryptocurrency thefts

Hackers utilize a variety of tactics to steal cryptocurrency from investment accounts. They can take cryptocurrency from investors' crypto wallets, build up fake wallets to bilk counterparties, and set up bogus crypto exchanges to steal

TABLE 5 Relative comparison of state-of-the-art crypto frauds on crypto romance scams.

Paper	Year	Key findings	Limitations and open issues
He et al.[51]	2021	Focused on the issue of fraudulent user identification in dating applications.	The proposed system cannot detect location-based attacks.
Suarez-Tangil et al.[50]	2019	Proposed an ML-based early detection method to catch romance fraudsters while they construct false accounts or connect with potential victims.	The suggested approach needs to notify and protect users from fraudsters at the local level. The system's efficacy in detecting romantic crypto fraud is not explored.
Cross et al.[49]	2018	Investigated the links between romance fraud and domestic violence and abuse.	Improved investigation of the real-time detection of crypto romance frauds is needed.
Whitty and Buchanan[47]	2016	Investigated the psychological impact of the online dating romance scam.	More research on the real-time detection of crypto romance frauds is necessary.
Whitty and Buchanan[48]	2014	Examined the online romance scams, in which fraudsters build electronically facilitated (false) emotional engagements with their victims to scam them.	Additional study into the real-time identification of crypto romantic scams is required.
Rege[46]	2009	Presented an overview of romance scams and identity fraud and then provided a preliminary evaluation of online criminal structure.	There is no practical application for identifying and investigating romantic crypto fraud.

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consumers' money.[52] Crypto-jacking is a major scam in the cryptocurrency realm. This is the unauthorized use of another person's computer to mine cryptocurrency. It is effectively hijacking another computer's processing speed, allowing all infected computers to mine BTC for the fraudster while also providing a way with larger profits and less danger of being caught. Figure 4 shows the amount of cryptocurrency theft during recent years.

3 | TAXONOMY OF CRYPTO FORENSICS

Economic crime and the trading of illegal items have gone online as the globe has grown more digital. The use of the dark web is increasing at a rate of 50% per year, and cryptocurrencies have replaced cash as the primary means of commerce. As a result, they have become essential facilitators of internet crime. We fight the rising use of pseudonymous and anonymous cryptocurrencies to promote illicit activities and financial crimes such as fraud, money laundering, and tax evasion at crypto forensics. Unlike fiat currencies, managed by a centralized body such as banks, cryptocurrencies are not managed by a single authority. In the case of conventional money, detectives will contact the appropriate authorities to seal the account and prevent further unlawful transactions. However, in the examination of cryptocurrencies, no centralized organization can be contacted to freeze the account. To solve this difficulty, studies[54,55] have indicated that an appropriate technique for analyzing cryptocurrencies is required.

Figure 5 shows the taxonomy of crypto forensics. Analyzing the various forms of cryptocurrency transactions and related frauds is the first step in the taxonomy of crypto forensics. Here are some examples of significant cryptocurrencies.

3.1 | Cryptocurrency transaction

3.1.1 | BTC transaction

In a blockchain system, transactions form the backbone of business operations. Transferring data between addresses is a critical aspect of a transaction's reliability. A typical BTC transaction includes pertinent data such as the addresses of both parties and the transaction value. The sender's address, also known as the input address, and the recipient's address, known as the output address, are included in the transaction. A transaction can contain several input and output addresses, and any remaining coins from the sender are stored in a designated address called the change address. Figure 6 demonstrates that a transaction's input address is derived from the output of a preceding transaction, which subsequently becomes the input for future transactions.[56] However, blockchain addresses' anonymity is not perfect, and malicious actors can use the flaws in the system to discern the link between a user's true identity and their

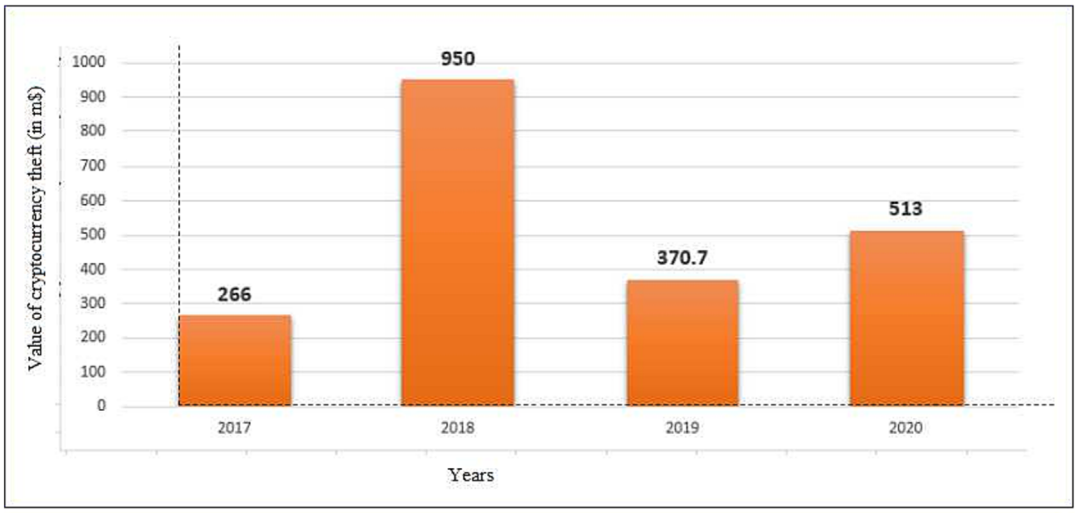


FIGURE 4 Value of cryptocurrency theft globally during 2017–2020.[53]

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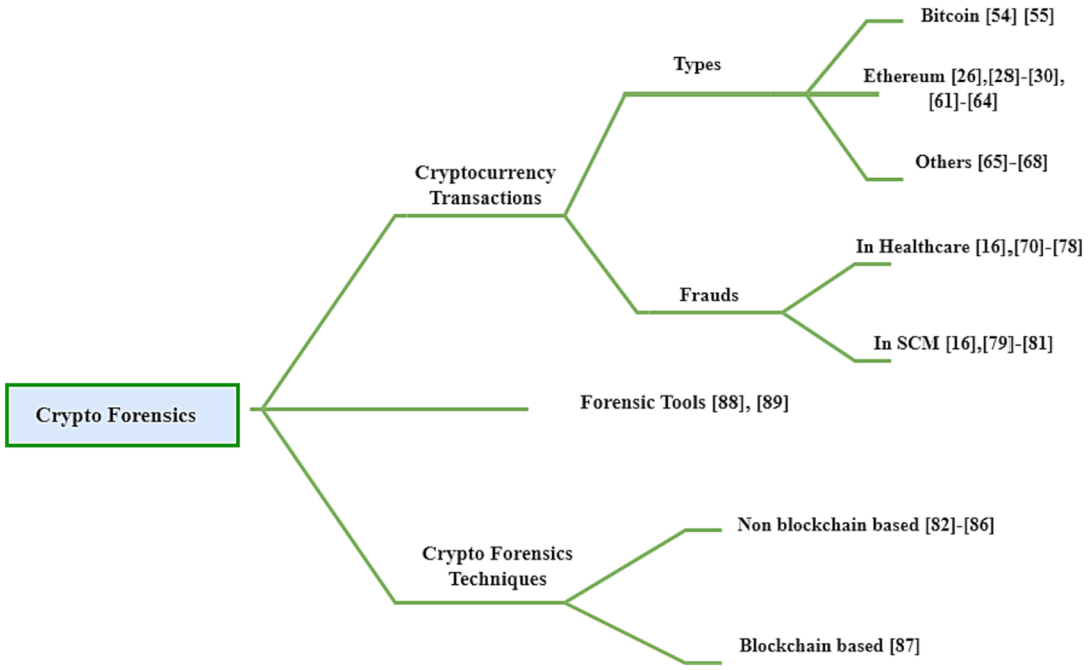


FIGURE 5 Taxonomy of crypto forensics.

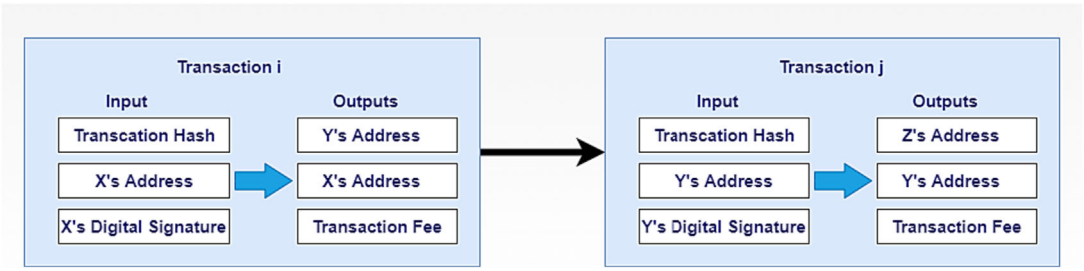


FIGURE 6 Transaction of Bitcoins (BTCs) from user X to Y and Y to Z

addresses.[57,58] There are certain common flaws, such as the address relation inside a transaction, the address relation across transactions, and the relationship between blockchain addresses and actual identities.

The analysis of BTC transactions in Meiklejohn et al[59] seeks to identify possible fraud. The researchers discovered that a few entities were responsible for a considerable portion of the network's transaction volume. Many of these entities indulged in suspicious activities, including using several accounts to conceal the source of money and their eventual destination. Vasek et al[60] examined the occurrence and effect of denial-of-service (DoS) attacks on BTC exchanges. Such attacks, the authors discovered, were prevalent and might be used to influence exchange rates or steal BTCs from unaware users. The authors of earlier studies[61,62] used a machine-learning-based technique to identify illicit activity on BTC transactions. K. Toyoda et al[63] has worked on immediately detecting HYIPs on the BTC blockchain.

3.1.2 | Ethereum transaction

Ethereum is a blockchain-based platform for developers to build decentralized apps (dApps) and smart contracts. Ether, the Ethereum network's cryptocurrency, performs transactions between external accounts. Among other things, transactions indicate the addresses of the source/destination, the Ether value, and a digital signature that validates the request.[64]

One popular type of smart contract scam includes fraudsters establishing a fake ICO or dApp and then employing a smart contract to mislead investors into paying Ethereum to a fake address.[28,30] proposed ponzi scheme detection

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techniques for Ethereum smart contracts. The other major type of crypto fraud in Ethereum cryptocurrency is fake Ethereum wallets. Fraudsters develop fake Ethereum wallets that seem authentic but aim to steal users' private keys and access their assets.[65] detects malicious nodes in account transactional activity using supervised ML-based anomaly detection.[66] introduces the Light Gradient Boosting Machine (LGBM) technique for identifying Ethereum fraudulent transactions. Ethereum exchange hacks occur when cybercriminals successfully gain illegal access to cryptocurrency exchange systems, stealing consumer funds. Hackers acquired access to the CoinDash website in 2017 and modified the Ethereum wallet address supplied on the site, forcing traders to unknowingly transmit money to the hackers' wallet.[67]

The authors in[32] first collect transaction records from Ethereum, then construct a transaction network from the transactions and propose a detection system based on graph convolutional network (GCN) to discriminate Ponzi contracts properly. After that, Chen, Weimin,et al.[31] proposed a semantic-aware detection method for noticing Ponzi scams in Ethereum. Wang, Lei, et al[30] projected a Ponzi scheme recognition technique for smart contracts based on LSTM. Jin, Jie, et al[28] proposed Ponzi-Warning, a dual-channel initial warning mechanism for Ethereum Ponzi schemes.

3.1.3 | Other cryptocurrencies

Because of their popularity and broad acceptance, cryptocurrencies with a large market value and trading volume, such as BTC and Ethereum, may be more vulnerable to fraud. Cryptocurrencies with low market size or those that are not well-regulated may also be more prone to fraud. Because of the lack of monitoring or regulation, these cryptocurrencies may be simpler for fraudsters to manipulate or exploit. Ripple is a cryptocurrency that wants to make cross-border payments easier by offering a faster and less expensive alternative to established methods. It works on a centralized network and uses the Ripple Protocol Consensus Algorithm for consensus. Ripple's native cryptocurrency is known as XRP. Like other cryptocurrencies, Ripple has been the target of giveaway scams. Scammers impersonate respectable firms or persons and promise huge sums of Ripple in return for a small initial payment.

Fraudsters have also constructed fake XRP wallet applications and websites that seem legitimate but are intended to steal users' XRP. Fake Ripple ICOs have also occurred in the same way as fraudulent Ethereum ICOs have.[68] The Securities and Exchange Commission (SEC) of the United States launched a lawsuit against Ripple in 2020, stating that it had executed an unregistered security offering by selling XRP tokens.[69] Tether is a stablecoin that aims to keep its Value stable about the US dollar. It's widely used in the cryptocurrency sector to move Value between exchanges or protect against other cryptocurrencies' volatility. On the other hand, Tether has been the topic of debate, with some critics claiming that it has insufficient reserves to back up its claims of being wholly collateralized. On November 21, 2017, Tether stated that it had been hacked and had lost $31 million in USDT from its primary treasury wallet.[70] Tether was also fined $42.5 million by the US Commodities Futures Trading Commission (CFTC) in October 2022 for misrepresenting its reserves.[71]

3.1.4 | Cryptocurrency fraud

Any fraudulent behavior utilizing cryptocurrency is referred to as cryptocurrency fraud. By facilitating and extending cybercrimes, the usage of BTC in various illicit activities undermines the broader threat picture. Darknet marketplaces, often known as cryptomarkets, are prime examples of how BTC intersects with illegal behavior. These unauthorized markets allow criminals to buy and sell a wide range and quantity of illegal things such as Weapons, ammunition, illegal wildlife parts, child pornography, contract killers, ransomware products, and much more which are available to buyers.[72]

1. In Healthcare: The National Health Care Anti-Fraud Organization (NHCAA) predicts that healthcare scams cause tens of billions of dollars in financial losses every year.[73] Scams in healthcare systems have not only resulted in higher costs but have also degraded the quality of treatment that patients should expect[7475]
   - Payment for fraudulent claims can be used to make payments for fraudulent health care insurance claims. Fraudsters can use BTC to receive payments for fictitious medical procedures, inflated medical bills, or unnecessary medical equipment. An Ohio physician was punished to 20 years in the central prison in December 2015 for performing unneeded catheterizations, testing, stent insertions, and inducing needless coronary artery bypass operations as part of a $29 million billing scam to Medicare and some other insurers.[73,76] A Virginia OB/GYN

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was arrested in 2019 and accused of performing unneeded procedures on female patients to collect insurance money.[73,77]

- Fraudsters sell counterfeit medical products such as pacemakers, thermometers, respirators, insulin pens, and hearing aids. Aortic pumps worth $7 million were recalled in the United States due to counterfeit components. The pumps appeared authentic, but they might malfunction and cause death.[78] Fraudsters who manufacture counterfeit medical supplies can use BTC to pay for their fake products in an untraceable manner. The usage of BTC can make it challenging to identify the culprits and recover the money.[79] It may also be more straightforward for counterfeiters to conceal their identities and avoid arrest due to the anonymity of BTC transactions.[18] The Silk Road, an online marketplace notorious for easing the sale of illegal products, particularly counterfeit drugs, is likely the most well-known Darknet.
- Since the internet has changed healthcare, several cyberattacks have targeted healthcare data.[80] Healthcare data have been stolen and kept for ransom in some cases. It entails hackers getting illegal access to healthcare systems or data and then encrypting the data so that it cannot be accessed without a decryption key. After gaining unauthorized access, the hackers demand payment in BTC to provide the decryption key. The pilfering of protected health information jeopardizes the safety and well-being of patients and health plan members, not just through identity theft and fraud but also by posing a severe threat to patient safety.[81]
- Additionally, fraudsters have leveraged the popularity of cryptocurrencies to mislead consumers into investing in fraudulent healthcare-related startups or organizations that claim to be developing new medical solutions. One form of fraud involves the sale of fraudulent "discount cards," claiming that consumers may use them to pay for prescription drugs or health care services.[82]

2. In Supply chain management:
   - In supply chain management, BTC may be used to purchase counterfeit goods. Because BTC is decentralized and anonymous, counterfeiters frequently utilize it to take payments, making it hard to recognize the criminals behind the transactions.[18,8384]
   - A ransomware attack occurs when a hacker infects and encrypts a company's computer systems. The hacker then wants BTC in exchange for the decryption key. If the organization refuses to pay, the hacker may threaten to expose sensitive data to the public, such as client information.[83] The COVID-19 pandemic aided the rise in ransomware attacks on supply chains by generating millions of possible entry points as businesses shifted to remote working.[85]
   - Fraudulent supply chain financing uses BTCs to get suppliers to provide products or services by promising them low-interest loans. This technique involves offering the suppliers a financing option that appears to be legitimate and calls for an upfront payment of a small fee, commonly via a fake website or email. The scammers disappear when the supplier pays the charge, leaving the supplier without funding and losing the advance payment.
   - Phishing schemes are used by fraudsters to steal private keys to BTC wallets. They establish fake websites and emails that mislead users into stealing their private keys. The fraudster may access the BTC wallet and transfer funds once they have the private keys.[86]

3.2 | Crypto forensics techniques

The second aspect of the taxonomy of crypto forensics is the technique used to investigate digital currencies. These techniques are classified into two main types: Non-blockchain analysis and blockchain analysis. The Non-blockchain analysis involves examining other data sources, such as social media, network analysis, exchange forensics, and cloud forensics, to identify the parties involved in a digital currency transaction. This technique can identify the IP addresses and social media accounts associated with a transaction. Network forensics studies network traffic to detect and analyze cyber-attacks and other security issues involving cryptocurrency.[87–89] Cloud forensics is the investigation of cloud-based storage and cryptocurrency-related services such as digital wallets and exchanges to trace the movement of funds and uncover suspicious activities.[90,91] Finally, blockchain analysis analyzes a cryptocurrency's blockchain to follow transactions and identify the individuals involved.[92] This category includes tracing the movement of funds and identifying suspicious activities by examining the blockchain, the public ledger that records all transactions in a specific cryptocurrency network.

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3.3 | Forensic tools

The third component of the crypto forensics taxonomy is the forensic tools used to study crypto forensics. These tools are classified into two main types: Open-source tools and Proprietary tools.[93,94] Open-source tools are software programs that are openly available and can be used to explore cryptocurrencies, for example, BTC Core and Electrum, while proprietary tools are software products that a company manages and must be purchased or licensed to use; Chainalysis and CipherTrace are two examples of proprietary tools.

Figure 7 shows the main process of crypto forensics. To properly investigate and convict organized cyber criminals, cryptocurrency forensics imposes strong collaboration between digital forensic experts and representatives of many institutions in the policing and judicial ecosystem. Crypto forensics software is vital in aiding law enforcement, intelligence agencies, regulatory bodies, and financial crime investigation teams. By leveraging advanced techniques, this software enables investigators to detect and uncover suspicious activity on the blockchain. The analytics platform that powers crypto forensics software draws insight from private data to assist in identifying suspects, demonstrating illegal activities, and gathering information on criminal businesses. Crypto wallets are crucial in investigating cryptocurrency cases because they are mathematically connected to the addresses used to perform transactions.[55] Several forensics investigations into cryptocurrency wallets are currently being done. Private keys, public keys, addresses, recovery phrases, wallet IDs, emails, wallet applications, file signatures, prefetch files, pagefile.sys, and fiberfill.sys are crucial artifacts for crypto wallets.[95] Finding evidence of cryptocurrency usage, such as transaction histories and targeted addresses, is the key objective of forensics. Crypto wallets installed on Windows systems have been studied by.[95,96] They initially acquired the RAM's contents to capture and preserve the data and then obtained data from the hard drive. After gathering the data, it was processed to find transaction histories and establish the existence of crypto wallets. L. Van Der Horst et al. and T. Volety et al[96,97] emphasized that even if the crypto wallet was retrieved and access was acquired, there is no assurance that the cryptocurrencies were correctly stored in the wallet. Seizing a crypto wallet may not guarantee its security because the wallet's private key or recovery phrase may be used to access it from other devices. Researchers such as T. Volety et al[97] have studied the security of various desktop crypto wallets, such as MultibitHD and Electrum, and identified flaws in their design. Similarly, L. Van Der Horst et al[96] have examined privacy-oriented crypto wallets like Monero and Verge installed on the Linux operating system and found that data like recovery phrases, public addresses, transaction IDs, wallet directories, and private keys can be recovered in encrypted form by imaging the RAM and disc. In a different field, R.U.Rahman et al[98] have presented a web forensic framework for investigating bot crime. They created a program to capture and save data from web application logs. Additionally, A. Al-Dhaqm et al[99] have proposed a methodology for database forensics investigation. In the Acquisition and

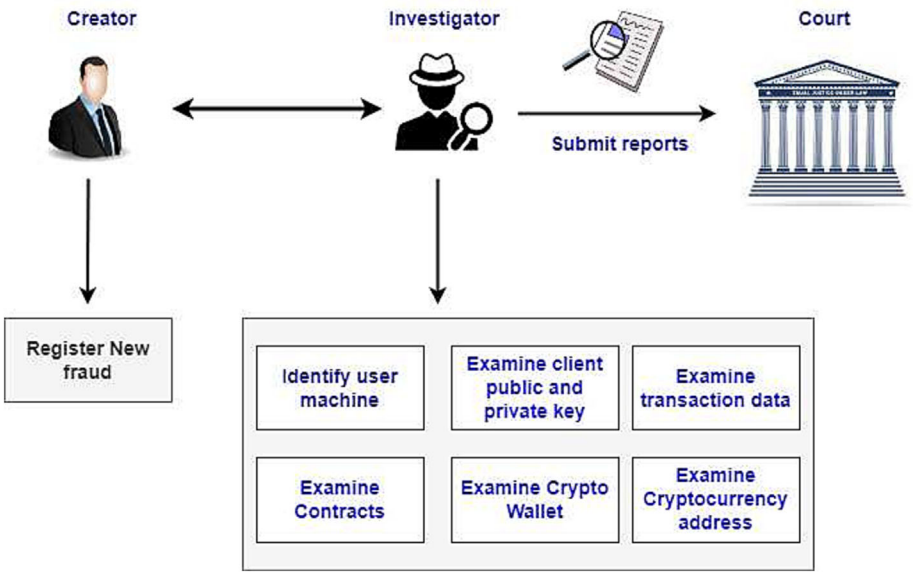


FIGURE 7 Process of crypto forensics.

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Preservation phase, data is obtained from the seized source and preserved using hashing, integrity checking, and backup techniques. This process involves extracting or imaging databases and hashing the data to ensure its integrity.

4 | USE OF BLOCKCHAIN FORENSICS TO INVESTIGATE CRYPTO FRAUDS

As the cryptocurrency market expands, so does the demand for people, organizations, and corporations to comprehend, track, and ascertain how and where cryptocurrencies are utilized. As blockchain technology is included, the digital forensics system is changing, making identifying those who have committed cybercrimes or cryptocurrency crimes easier. Blockchain forensics is a revolutionary way to punish fraudsters quickly. It is a new sophisticated technique that uses science and technology to fight fraudsters in the cryptocurrency industry. The primary purpose of Blockchain Forensics is to recover and evaluate various types of evidence that have been left on the most beneficial and transparent blockchain digital leader. With their unique skill set, expertise, and continuous education, forensic accountants and blockchain and digital forensic specialists are uniquely positioned to aid in tracking crypto assets, providing clarity, and conducting a complete investigation. Most cryptocurrency transactions, including BTC, Ethereum, and others, are private. There is a prevalent misperception that blockchain technology ensures perfect anonymity for users. This is accurate because a user's wallet address is not tied to a real-world identity. Blockchains are distributed ledgers in which the database is maintained on each node in the network. This increases openness by allowing anybody to view the status of a database and the transactions that take place on it. To assist in narrowing down a network user's identity, agencies might track the trail of transactions on the blockchain. We can examine each circumstance using specific blockchain forensics tools to assist in discovering the entrance and exit points of BTC transactions. We cannot eliminate fraud and money laundering, but we can limit it using forensic techniques.

Blockchain Explorer, BTC Block Explorer, Wallet Explorer, Blockparser, ORS CryptoHound, and many other popular tools for tracking individual transactions are available. Various blockchain intelligence businesses, such as CipherBlade, have risen to prominence in current years as the number of crimes in this area has skyrocketed. Chain analysis, a blockchain intelligence business, has obtained multi-million-dollar contracts from the US government for blockchain analysis and surveillance in recent years. MasterCard bought CipherTrace, a company that examines blockchains for illegal transactions. GraphSense is an open-source program supporting several cryptocurrencies, including BTC, Cash, Zcash, Litecoin, and Ethereum.[12] It is a powerful tool for tracking monetary flows, offering a comprehensive transaction history, and enabling visual analysis through a user-friendly web interface is available. EthExplorer specializes in examining the Ethereum blockchain and smart contracts, facilitating path and graph pattern search for identifying abnormal activity between crypto addresses. BlockTag is tailored to work as a tagging system that links items with real-world identities, while BlockSci is a quick programming interface that allows users to run customized queries on the blockchain.

4.1 | Main functions of blockchain forensics tools

Blockchain forensic tools serve several key purposes for understanding and analyzing blockchain transactions.

- Address Classification: Associating blockchain addresses with real-world identities is one of the most important uses of blockchain analysis tools. This critical task requires specialized techniques to identify the owners of digital wallets and connect them to physical individuals or organizations.
- Transaction monitoring and risk assessment: By keeping a close eye on every transaction involving a company, investigators can evaluate the level of risk based on factors such as the source of the funds, the movement of the money, and the history of the wallets used by the sender and recipient.
- Provide visual aids: To aid in the manual investigation of transactions and addresses, forensic tools also provide visual aids. These investigation tools allow investigators to visualize the flow of funds and the relationships between different addresses, helping to uncover patterns and identify potential illicit activity.

One of the most prominent applications for blockchain forensic tools is the ability to correlate blockchain addresses to real-world identities. Such tools detect real-world items on the blockchain in several ways. Crypto trading bots, exchanges, and other services process billions of crypto transactions daily while also being hacked occasionally. To

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uncover the real identities behind blockchain addresses, blockchain analysis tools utilize techniques such as clustering algorithms, scraping data from websites, monitoring scam databases, and launching dust assaults. These methods enable the software to delve deeper into the blockchain and unveil any suspicious behavior that may have been previously hidden. Clustering is the most often used approach in analytic tools for identifying entities like exchanges, payment processors, and wallets. Web scraping is also used to examine the cryptocurrency market. It can keep track of price changes and save them in your database for later use. As a consequence, we'll be able to respond quickly once prices reach a certain level. Organizations that want to achieve security and compliance for their cryptocurrency businesses can now use cryptocurrency investigation software and educate their employees on compliance-related regulations such as the FATF guidelines on transaction monitoring, suspicious activity reporting, and responding to risky activity. Decentralized exchanges and decentralized financial institutions will benefit from cryptocurrency-compliant software in understanding and complying with local legislation. Blockchain analysts can help compliance teams navigate the most complicated investigations and build a strong anti-fraud program.

Blockchain risk analysis is performed by developing and training risk models via ML techniques such as clustering to assign a risk score to each blockchain transaction. Multiple parameters, such as the source of funds, amounts of transactions, the destination of funds, and money flow history, are utilized to develop risk models. Anti-fraud crypto monitoring systems analyze publicly available transaction data to detect unlawful and criminal activity. When suspicious behavior is spotted, the wallet or transaction is awarded a risk score, inspected for its specified risk level, and accepted, producing an end-to-end trail. Data collection and transaction identification in the crypto compliance tool are ongoing processes. Analytics operates as the front line of defense in ensuring that fair and lawful procedures are followed, contributing to the cleanliness of crypto ecosystems. The software for analyzing blockchain also comes equipped with an array of visualization tools that aid in detecting potential crypto risks. These tools allow for the investigation of blockchain addresses using transaction graphs, which provide insight into the connections between multiple transactions, helping to identify any suspicious activity. Furthermore, crypto compliance solutions record each transaction. Data modeling is also part of the blockchain surveillance capability since it contributes to the transparency and compliance of the cryptocurrency markets.

Graph mining algorithms were proposed in Phetsouvanh et al[100] to investigate the links between BTC wallet addresses. Neighborhood subgraphs are analyzed in terms of path length and convergence to discover unusual BTC movement and other wallet addresses held by the suspected culprits.[22] projected an identification methodology based on unit conversion and a sampling strategy to identify whether a certain BTC address is managed by an HYIP. A. G. Khan et al[101] created a QR code-based Android application with a protected private key storage to preserve privacy and eliminate fraudulent attacker logins. Zheng et al[56] utilized a two-step clustering technique to associate BTC addresses, initially linking addresses by analyzing different inputs and currency change regulations and applying the enhanced Louvain algorithm to separate transaction addresses into multiple communities. Through this approach, the researchers identified a group of CryptoLocker blackmail addresses. Zhang et al[102] examined the hidden connections between various addresses in a BTC blockchain through transactions that might help disclose the true identities of the persons who own the addresses. Using BTC address clustering techniques, Li, Zhen[103] discovered the addresses of ten well-known cryptocurrency exchanges, including Huobi and Binance, and categorized the addresses into user addresses, hot wallets, or cold wallets. Aquilina, Simon Joseph et al[64] examined the Ethereum attacks using Indicators of Compromise (IoC) and developed a model to compare IoCs at various levels using a prototype tool EtherClue. Lamprini et al[104] implemented an Ethereum-based forensic investigation framework by preserving the chain of custody and uniform forensic flows for fraud investigations.

Decentralized infrastructure and platforms are required for creating decentralized programs in DeFi. DeFi enables consumers to freely engage in sophisticated blockchain transactions using contracts at a minimal cost. However, due to the versatility of DeFi, some dangers are unavoidably added. B. Wang et al[105] developed a real-time threat detection tool BLOCKEYE for Ethereum-based DeFi. This tool worked in two phases. In the initial phase, a symbolic execution analysis was conducted using Oracle to detect whether external manipulation of state data streams had occurred to identify vulnerable DeFi systems. The second phase involved collecting transactions through transaction monitors beneath the chain, extracting features, and conducting a more in-depth attack analysis. Mun et al[106] presented a framework for analyzing BTC data using SQL. They reduced database size by employing data mapping and transaction splitting techniques. Then they investigated address management behaviors in exchanges using taint analysis, which follows BTCs back to the user's wallet. Table 6 demonstrates the relative work on Blockchain and Crypto Forensics for crypto frauds.

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TABLE 6 Related Work on blockchain and crypto forensics for crypto frauds.

Paper	Year	Key findings	Limitations and Open issues
Mun and Lee[106]	2022	Developed a compact database for BTC analysis by data mapping approach and a split transaction mechanism.	This study is exclusively concerned with the storage of BTC data. Not dealing with the investigation and prevention of cryptocurrency-related frauds.
Taneska et al.[5]	2022	Presented a forensic review of certain privacy-oriented cryptocurrencies', blockchain networks, and wallets.	The chances of the forensic investigation gaining access to the criminal's wallet are quite low.
Wang et al.[105]	2021	Proposed a real-time threat detection tool, BLOCKEYE, for Ethereum-based DeFi.	It isn't easy to assess the success of the suggested technique on systems other than the Ethereum blockchain network.
Zarpala and Casino[104]	2021	Proposed an Ethereum-based forensic investigation framework based on standardized protocols and the associated methodology for fraud scheme investigations.	This study does not extend beyond the Ethereum blockchain network, and the suggested approach cannot automatically identify other types of crypto fraud.
Aquilina et al.[64]	2021	Examined the various Ethereum attacks using Indicators of Compromise (IoC) and suggested a paradigm for smart contract execution.	The proposed technique and tool are not entirely capable of automatically detecting other crypto frauds in blockchain networks other than Ethereum.
Li et al.[103]	2020	Presented a method for tagging BTC addresses of exchanges using BTC address clustering algorithms and categorized such addresses as user addresses and hot or cold wallets.	Some exchanges have distinct behavioral patterns. All of these exchanges are handled unevenly by the suggested approach.
Zheng et al.[56]	2020	Developed a heuristic clustering approach for judging the relationship between BTC addresses and applied the Louvain method to determine the relationship between BTC users.	It is hard to determine the effectiveness of the suggested clustering heuristic approach.
Zhang et al.[102]	2020	Proposed a heuristic approach that identified one-time alteration addresses by deleting reused or non-change addresses.	It is hard to assess the quality of the projected heuristic due to the absence of a comprehensive set of ground-truth labels, which could help determine the accuracy of the clustering algorithm at a large scale.
Khan et al.[101]	2019	Proposed two Android applications and added numerous signatures and keys to the original wallet structure for interactive authentication.	The security issues associated with QR code scanning are not addressed. Because of advancements in attacker strategies, the suggested solution is insufficient.
Toyoda et al.[22]	2019	Proposed a HYIP identification methodology based on unit conversion and a sampling strategy.	The suggested classifier's capability is confined to detecting BTC-related fraud.
Phetsouvanh et al.[100]	2018	Proposed graph mining algorithms to investigate the links between wallet addresses.	This article focused on detecting/aggregating attackers' BTC wallet addresses. Other cryptocurrencies should be investigated.

5 | AI AND BLOCKCHAIN-DRIVEN ARCHITECTURE TO DETECT CRYPTO FRAUDS: A CASE STUDY

The projected framework is a system that uses AI and blockchain technology to detect and prevent cryptocurrency fraud. The architecture has been divided into two primary parts: AI and blockchain, which work together to detect and record instances of fraud. It uses numerous ML classification algorithms in the AI section to detect and categorize fraudulent activities in the cryptocurrency market. These algorithms analyze data from a variety of sources. ML Classifiers are trained on past data and constantly learn from new data to improve accuracy. When they detect possible fraud, the data are securely saved using the IPFS, a distributed file storage system built for decentralized and peer-to-peer networks. It assures that data cannot be tampered with or altered by a single entity. The file's hash is then saved in the blockchain using smart contracts.

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Figure 8 shows the complete proposed system model, which comprises three layers, i.e., cognitive, AI, and protection. A dataset including many crypto transactions from various sources is available to the cognitive layer. After the cognitive layer prepares the dataset, it is input into the AI layer. The AI layer then uses this data to train the neural network architecture, enabling it to make predictions on raw, unprocessed data with accuracy and efficiency. Finally, when the ML classifier identifies fraud, it triggers the protection layer, which alerts law enforcement authorities and provides them with transaction and wallet details via the blockchain smart contract.

The following is a detailed explanation of the system:

- Cognitive layer: The cognitive layer retrieves cryptocurrency transactions from diverse sources, including online marketplaces connecting buyers and sellers of various goods and services, freelance service platforms linking clients with freelancers for different services, and financial trading platforms enabling users to trade cryptocurrency and other financial instruments, among others. An example of the potential use case for cryptocurrency transactions that span multiple application areas is the trade of human organs using cryptocurrency. This scenario involves buyers and sellers from different regions who connect through a cryptocurrency marketplace platform. The buyers may be individuals or organizations seeking organs for transplant purposes, while the sellers could be individuals or groups offering to sell their organs in exchange for financial compensation. However, there is a risk of fraudulent activity in this marketplace, such as price manipulation or misrepresentation of the quality or condition of the sold organs. So, cryptocurrency transactions are stored in a database and processed for optimal output. The raw dataset is then converted into a CSV format compatible with ML. According to the ML perspective, the task is a binary classification problem, with normal data assigned a value of 0 and fraudulent data assigned a value of 1. Because of the existence of outliers that might mislead classifiers, the raw dataset requires preprocessing to accomplish this classification. If the data is

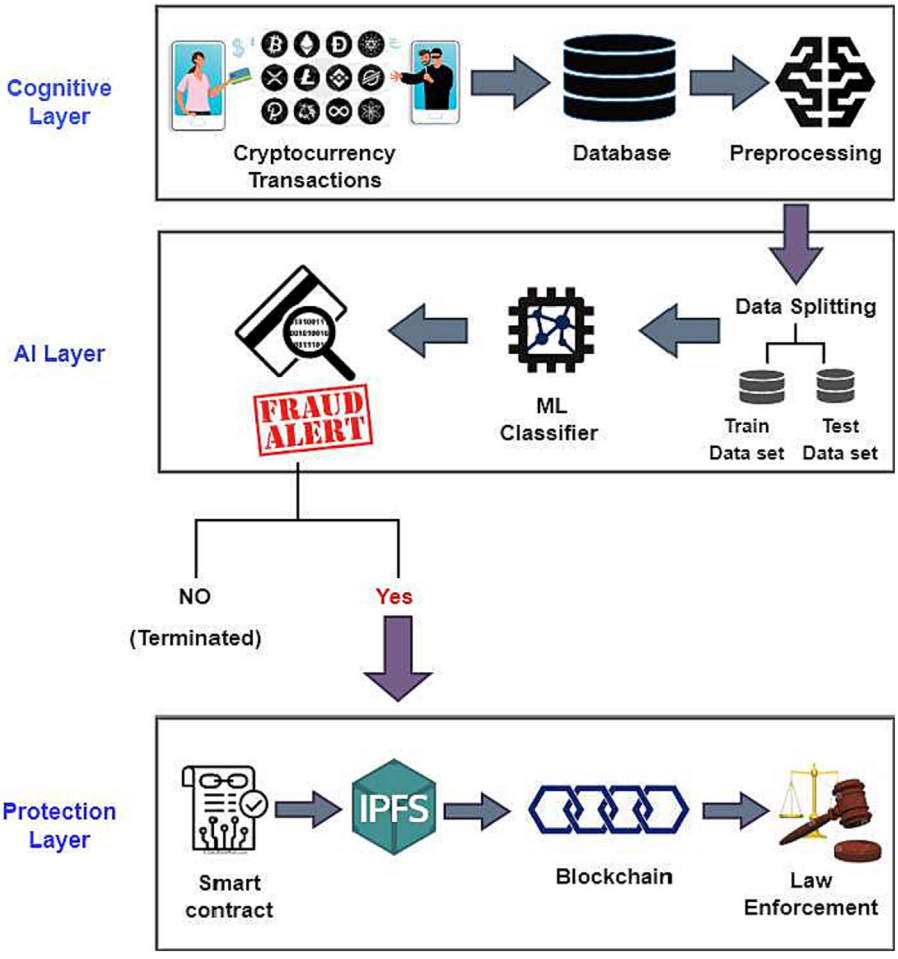


FIGURE 8 AI and blockchain-driven architecture to detect crypto frauds.

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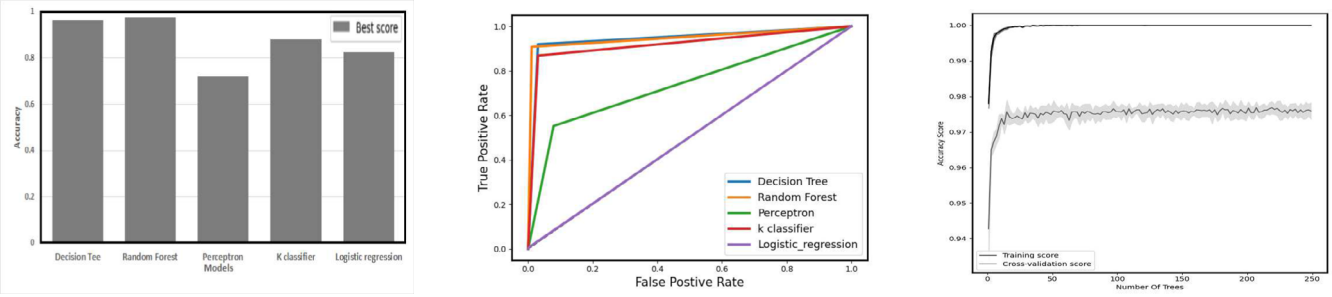
not normalized, the columns may contain small and large values, and missing values may cause bias in the learning model. Thus, the dataset is preprocessed using various techniques, including filling in missing values, outlier detection, normalization, and feature selection. The dataset is also checked for imbalance, where the presence of a majority class can lead to bias in the ML model. To overcome this, resampling techniques like oversampling and undersampling are applied to balance the dataset. To validate the final prediction, the balanced dataset is divided into training and testing data sets.

Dataset description

Detecting crypto fraud poses a significant challenge primarily due to the scarcity of publicly accessible datasets, stemming from concerns surrounding privacy and the potential inclusion of sensitive information. For our study, we sourced our dataset from the Kaggle repository, which consists of genuine Ethereum blockchain transactions. These transactions are categorized into two groups: known fraud and legitimate transactions. Each entry corresponds to a transaction record associated with an Ethereum account. The dataset is characterized by 48 distinct features and encompasses a total of 9841 transactions. Each observation represents a unique Ethereum account, with each variable representing an aggregate statistic derived from all transactions conducted by that particular account. These statistics include metrics such as total Ether value received and average time intervals between transactions. Additionally, the dataset distinguishes between transactions made between accounts and those involving smart contracts. However, it is worth noting that the dataset exhibited a notable imbalance, with only 22.14% of the entries classified as fraudulent. To address this, we initiated a data cleansing process by removing insignificant characteristics with missing values or those deemed irrelevant to our model. Subsequently, we selected 7872 out of the 9841 transactions for training purposes, while the remaining 1969 were reserved for testing. Leveraging this dataset, we proceeded to train a model tasked with determining the fraudulent nature of the provided transaction data. It is important to acknowledge that further augmentations to the dataset may yield improved accuracy, potentially leading to enhanced outcomes in future applications.

- *AI layer:* In this layer, an ML classifier is utilized to identify fraud in data obtained from the cognitive layer. ML classifiers are algorithms that can automatically learn from input data and predict new data based on that learning. They are widely used in fraud detection and personalized recommendations. In fraud detection, ML classifiers are commonly used to identify suspicious patterns or actions that suggest fraudulent behavior. For example, classifiers may learn to recognize patterns and features characteristic of fraudulent transactions by examining large amounts of transactional data, such as unexpected purchase amounts, abnormal transaction frequency, or deviations from a customer's normal behavior. So, in the AI layer, the analyzed data are separated into two datasets: training and testing, and the training datasets are cross-validated with the test datasets. It is then passed through several classifiers, including Decision tree, RF, Perceptron, k-classifier, and Logistic regression, to differentiatex'x' between legitimate and fraudulent transactions. Figure 9 describes several tests using these classifiers. Figure 9A illustrates the accuracy chart. This is a visual representation of the performance of different ML classifiers. From the accuracy values in the chart, we find that the random forest classifier had the highest accuracy, followed by the decision tree and KNN classifiers, while the perceptron and logistic regression classifiers had lower accuracies.

  Figure 9b shows the Receiver Operating Characteristic (ROC) curve, which is a plot of the true positive rate (TPR) against the false positive rate (FPR) at diverse threshold values, with the area under the curve (AUC) representing the performance of the classifier. As we can see from the ROC curves, the random forest classifier has the highest



(A) Accuracy comparison. **(B)** ROC-AUC curve **(C)** Validation curve

FIGURE 9 (A–C) Comparison of machine learning (ML) classifiers on different evaluation metrics.

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area under the curve (AUC), which specifies that it has the best overall performance among the classifiers. The decision tree and KNN classifiers also have relatively high AUC values, indicating good performance. On the other hand, the perceptron and logistic regression classifiers have lower AUC values, indicating poorer performance. So, Figure 9A,B revealed that the RF model provides a good combination of classification accuracy and range under the curve (AUC) score. Figure 9C shows the validation curve of the RF Model. In this curve, the number of trees is a hyperparameter that can significantly affect the model's performance. This validation curve shows how the model's accuracy changes as the number of trees in the forest increases.

- *Protection Layer:* The trained model can predict whether or not the transaction made by the user layer is fraudulent. If the model determines that the transaction is a fraud, the transaction address, wallet address, and other information are crucial for judicial officers. This information is susceptible, and if compromised as part of the fraud, it can be altered. This highlights the issue of data integrity. Therefore, a system for securely storing sensitive data is mandatory. Blockchain has the potential to be an answer to such a situation. So, Smart contracts validate the data before it is stored on the blockchain. They are software applications that run when specific parameters are met. Thus, all fraud transactions are stored in the IPFS network, and the corresponding hash value is kept in the blockchain network. Under the proposed architecture, only authorized persons would access the evidence on the blockchain. Once the data have been confirmed, it is securely kept within the blockchain, ensuring it cannot be changed later. Investigators can leverage blockchain technology in the suggested architecture to acquire safe access to information on crypto scams.

Several criteria for evaluation, including precision, accuracy, and F1-score, are used to evaluate the performance of the AI systems.[107] For example, Table 7 shows the accuracies of different CNN architectures.

5.1 | Evaluation metrics

Evaluation metrics enable us to compare different models rapidly and gain insight into how well they are trained on data and their performance on accurate data. The selection of the final ML classifier relies on various metrics, which are discussed in detail below. By examining these metrics, we can make informed decisions about which model is best suited to our needs.

5.1.1 | Confusion matrix

A confusion matrix is a tool used to visualize and summarize the performance of a classification algorithm. It comprises four essential components that define the classifier's evaluation metrics, including the following:

- The number of correctly classified positive labels (true positive (τ)).
- The number of negative labels erroneously classified as positive (false positive(ϕ)).
- The number of correctly classified negative labels (true negative (υ)).
- The number of positive labels is erroneously classified as negative (false negative (χ)).

TABLE 7 Evaluation metrics (in %.)

Architecture	Accuracy	Precision	Recall	F1 score
DecisionTree	96.4	98.0	98.0	98.0
RF	97.5	98.0	99.0	98.0
Perceptron	72.0	78.0	88.0	83.0
KNeighbor	88.1	92.0	93.0	92.0
LogisticRegression	82.8	82.0	100	90.0

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By providing an overview of these four key metrics, the confusion matrix allows for a more comprehensive evaluation of the classifier's performance. Figure 10 shows the confusion matrix with the RF classifier.

5.1.2 | Accuracy

Accuracy is a significant statistic for measuring classification performance, well-defined as the proportion of correctly predicted events to all samples under examination. The following is a mathematical definition of accuracy.

$$Accuracy = \frac{\tau + \upsilon}{\tau + \upsilon + \phi + \chi} \tag{1}$$

Equation (1) shows the proportion of properly detected samples to all samples. Accuracy states how effectively a model executes, and it also helps in comparing diverse models.

5.1.3 | Precision

Precision is the proportion of accurately recognized positive samples to all positive predicted samples. Improved accuracy ensures that the proposed model produces a more significant number of accurate warnings. Precision is represented mathematically by the Equation (2)

$$\rho = \frac{\tau}{\tau + \phi'} \tag{2}$$

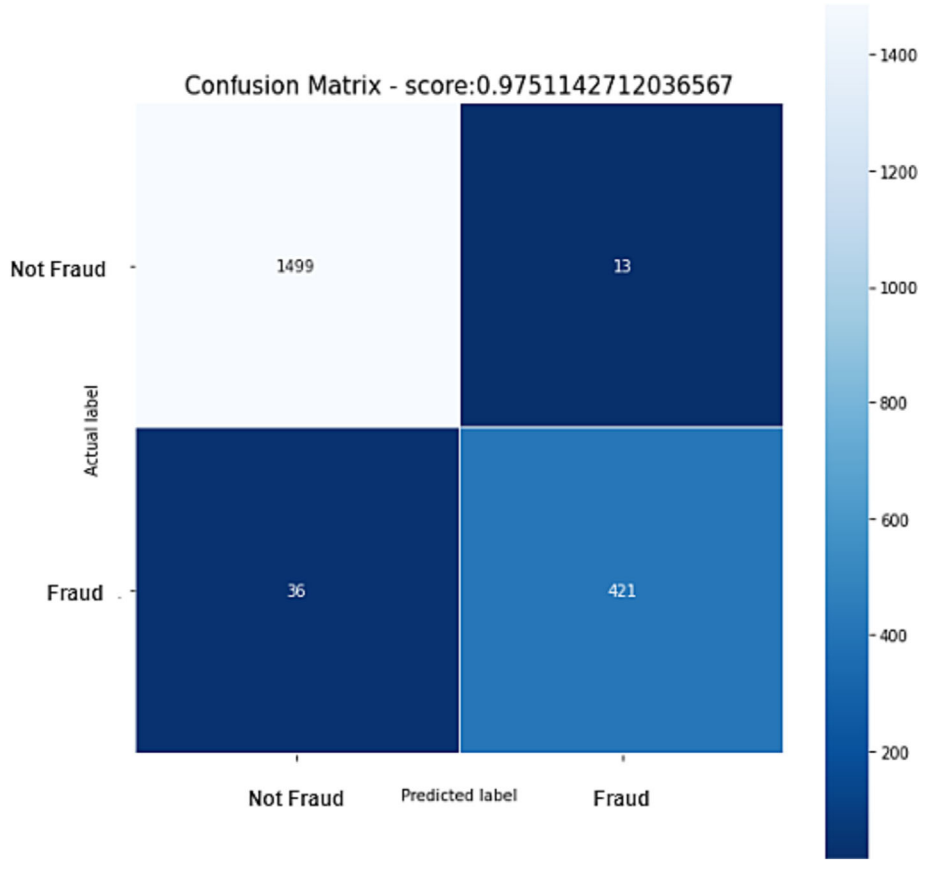


FIGURE 10 Confusion matrix for the random forest on the test dataset.

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where ρ denotes precision. Low precision shows that the model generates more false alerts in this context.

5.1.4 | Recall

The percentage of successfully identified positive samples to all positive samples is shown by a recall classifier. It is evaluated using the Equation (3)

$$\gamma = \frac{\tau}{\tau + \chi'} \tag{3}$$

where γ signifies recall. In this instance, a model with a high recall rate will produce more accurate warnings.

5.1.5 | F1 score

In cases where two models have identical accuracy and recall scores, the F1 score is a crucial metric. Its primary objective is to provide a more reliable means of evaluating the model when precision and recall scores are insufficient. The F1 score considers precision and recall, allowing a more comprehensive assessment of the model's performance. Furthermore, it was designed to function well with imbalanced data. As indicated in Equation (4), the harmonic mean of recall and accuracy is used to define the F1 score.

$$F1 - score = \frac{2 * \rho * \gamma}{\rho + \gamma} \tag{4}$$

The table shows that the RF exhibits the greatest F1 score, therefore being the final model selected.

5.1.6 | ROC curve

The ROC-AUC curve, representing the relationship between true positive rate (TPR) and false positive rate (FPR), is a valuable two-dimensional tool for evaluating model performance. The higher the AUC score, the more influential the model distinguishes between true and false positives. Figure 9B displays the ROC curve.

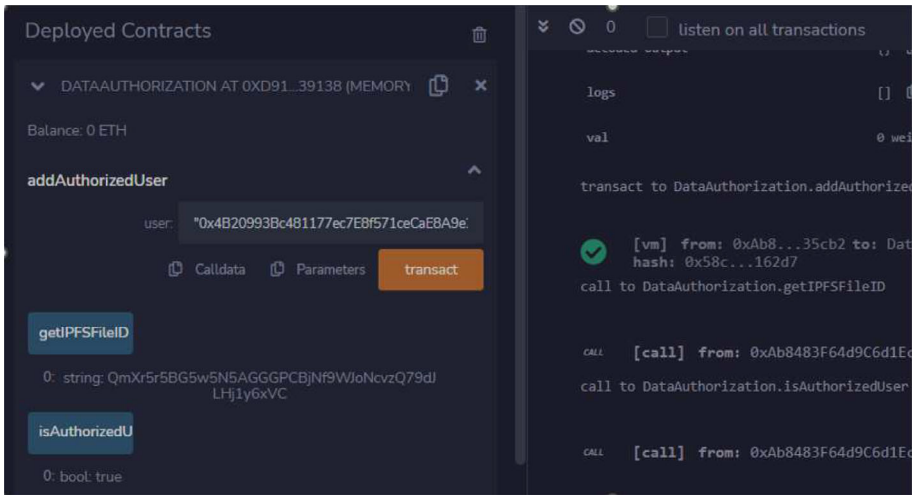
5.1.7 | Validation curve

A graphical method for measuring the influence of a single hyperparameter is the validation curve. By inspecting this curve, we may find whether the model is right for a particular set of hyperparameter values, underfitting, or overfitting. For example, Figure 9C shows the validation curve with RF.

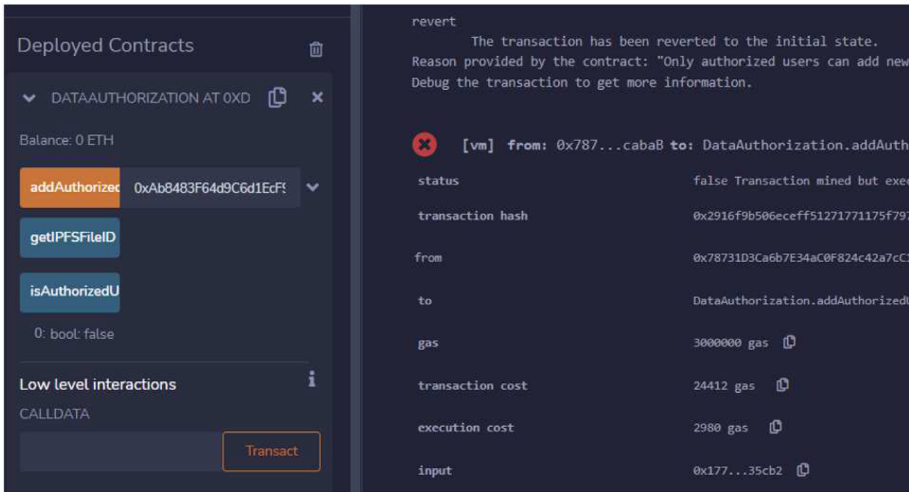
5.2 | Implementation of smart contract

The smart contract has been developed in Remix IDE using the solidity programming language, as illustrated in Figure 11. The addAuthorizedUser, getIPFSFileID, and isAuthorizedUser functions are included in the smart contract, as shown in Figure 11A. The addAuthorizedUser function allows authorized users to add new users to the list of authorized users. The isAuthorizedUser function checks whether a user is permitted, such as a police officer or an unwelcome intruder. Finally, the getIPFSFileID function enables the retrieval of the IPFS file ID, which can be used to access fraudulent transaction details. Furthermore, Figure 11B depicts that in the event of an unauthorized attempt to access the secure data stored on the blockchain, a revert error message will be triggered. As a result, the blockchain will be restored to its original state, effectively thwarting the attacker's efforts to gain access to sensitive information.

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(A) Data retrieved by an authorized official.



(B) Cautionary alert for an unauthorized user

FIGURE 11 Prototype model for finding details of the Fraudulent details.

5.3 | Bandwidth utilization

To simulate the bandwidth utilization of the smart contract, we use its hash code as input for IPFS. The hash code is a unique identifier representing the contract data. IPFS retrieves the contract from the network using this hash code and simulates the bandwidth utilization required to upload and download the contract. The network bandwidth utilization, measured in kilobits per second (Kb/s), is depicted in the graph generated from the IPFS simulations on a local machine. To run this simulation on our local system, we installed and configured an IPFS node. As a result, the bandwidth consumption is relatively low, as shown in Figure 12.

6 | CHALLENGES

Due to blockchain technology's decentralized feature and anonymity in cryptocurrency-related transactions, blockchain forensics and crypto forensics are challenging. Figure 13 depicts various challenges Blockchain Forensics and Crypto Forensics encounter while investigating cryptocurrency fraud.

1. *Challenges to investigate Crypto frauds:* Following are some fundamental obstacles for investigators when investigating cryptocurrency fraud:

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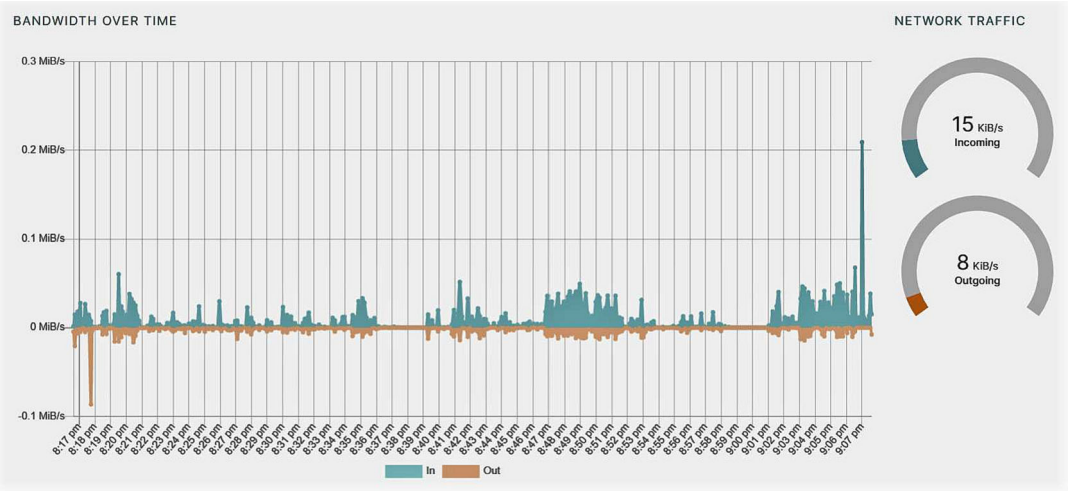


FIGURE 12 Bandwidth usage with InterPlanetary File System (IPFS) simulations.

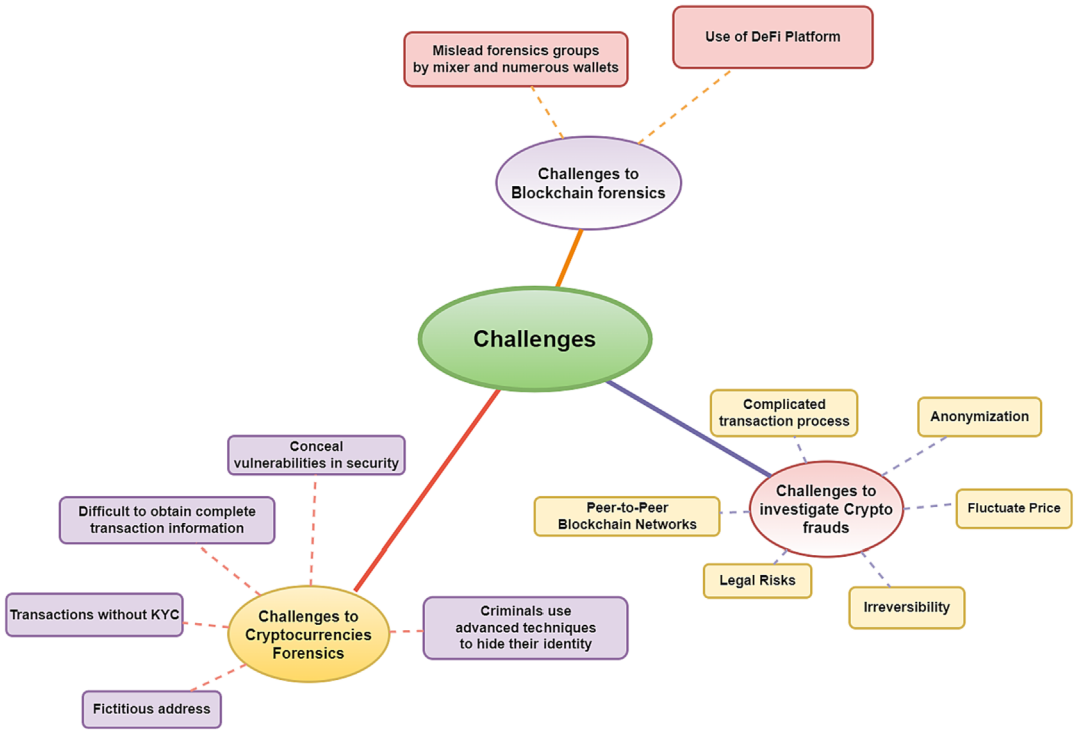


FIGURE 13 Challenges of investigating cryptocurrency frauds.

- Complicated transaction process: The main difficulty for the authorities recognizes the technological procedure throughout the transaction operations for each cryptocurrency algorithm.
- Anonymization: Several cryptocurrencies are currently implementing anonymization layers that conceal both the sender and receiver and the transaction cost. This anonymization will make it more difficult to detect and investigate various offenses.
- Peer-to-Peer Blockchain Networks: P2P decentralized networks enable users to swap cryptocurrencies or crypto assets without the intervention of a cryptocurrency exchange that requires KYC.
- Legal Risks: There are no legal safeguards for cryptocurrency payments. The growing NFT industry may become a target for money launderers and terrorist financiers looking to "clean" unlawfully obtained funds.
- Irreversibility: Cryptocurrency payments are often irreversible.

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- Fluctuate Price: Cryptocurrency prices are continuously changing.

2. *Challenges to cryptocurrencies forensics:* Following are the challenges of cryptocurrency forensics:
   - Conceal vulnerabilities in security: The fundamental challenge many organizations face nowadays is maintaining their image as a "trustworthy firm" capable of preserving its assets and being seen as trustworthy by its partners and consumers. As a result, even if cybercriminals target a business, management's first concern is to ensure that the security issue is not made public. This scenario is beneficial to cyber-criminals.
   - Difficult to obtain complete transaction information: The second challenge of cryptocurrency forensic investigation is gaining access to the complete information, including online segments.
   - Criminals use advanced techniques to hide their identity: Criminals increasingly employ advanced anti-forensic measures, such as utilizing a VPN (Virtual Private Network) connection to conceal their identity, even though their connection IP address is shared with law enforcement by their email service provider. Furthermore, they hired a VPN company that does not keep logs.
   - Transactions without KYC: A nested exchange (an OTC broker) maintains accounts on many significant cryptocurrency platforms and allows customers to trade via those accounts. The nested exchange often provides quick access to all functionalities without needing KYC. However, these exchangers provide money laundering services to scammers and fraudsters in many situations and may be leveraged for ransomware payments.
   - Fictitious address: Cryptocurrency such as Monero encodes the receiver's address on the blockchain and produces fictitious addresses to conceal the true identity. It also obscures the value of the transaction, making it effectively impossible for money laundering investigators to "follow the money."
3. *Challenges to blockchain forensics*: Following are the challenges of Blockchain forensics:
   - Mislead forensics groups by mixer and numerous wallets: Experienced criminals, or occasionally competent crypto investors, regularly try to mislead forensics groups or crypto monitoring systems by adopting a range of tactics and strategies to limit blockchain analysis, such as the use of mixers, numerous wallets, and non-compliant exchanges. One of the most common techniques is to use mixers to aggregate numerous senders' inputs while concealing their addresses. All other addresses are untraceable since the mixed input is transmitted from a single address, masking the original sender. A mixer (or tumbler) is a service that combines multiple streams of possibly traceable crypto assets. This improves transaction anonymity by making crypto transactions more difficult to monitor. Similarly, several wallets are formed between successive transactions to make law enforcement's investigation more complex. Additionally, utilizing noncompliant exchanges to evade KYC standards allows fraudsters or criminals to avoid blockchain examination.
   - Use of DeFi Platform: Digital Finance(DeFi) is a blockchain-based ecosystem for built financial applications. DeFi's distinguishing characteristic is that it allows members to lend, sell, and borrow money among themselves without the intervention of conventional financial institutions like banks. DeFi is unregulated, and there is no necessity to give KYC information for anti-money laundering purposes. In many situations, hackers exploit flaws in freshly built DeFi systems to steal money from DeFi protocol users.

7 | CONCLUSIONS AND FUTURE SCOPE

In the present research, we have conducted a thorough investigation of the field of Blockchain Forensics and Crypto Forensics to detect crypto frauds. Our analysis is based on advancements in traditional methodologies from year to year and an analytical examination of numerous advanced crypto frauds. We have presented a taxonomical classification of crypto forensics and blockchain forensics. Moreover, we suggest a framework incorporating AI and Blockchain to investigate crypto frauds securely. We analyzed the suggested framework using the case study based on cryptocurrency dealings from diverse sources. Several ML classification techniques have been applied to evaluate the efficacy of the suggested framework. Our study also highlights the remaining issues and research obstacles for investigating crypto fraud. Through our investigation of crypto fraud's attack mechanism and behavior, we have discovered that the attacker's information can be exposed through blockchain and crypto forensics. With this perspective, our research will significantly contribute to the fight against this challenge.

In the future, we will implement the survey observations to develop a unique solution to confront modern security attacks in cryptocurrency transactions by combining blockchain technology and AI models.

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DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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